

Effectiveness of Preventive Measures on Spread of Novel Disease in a Small Community

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Abstract

With the possibility of an disease outbreak happening it is important to model and predict the impact on a local, national, or global scale. The SIR-model can mathematically predict the fraction of the population infected over time but it is difficult to predict influences on parameters as an effect of using preventive measures. This paper extends this model by implementing change in behaviour of the 3 groups. Knowing more accurately what effects preventive measures have on the spread can greatly help in keeping the number of infected people under a controllable limit. The simulation has an area with an even distribution of people and a probability to infect each other on contact. The influence is measured by tracking the amount of people being healthy, sick, and immune over time. It is found that when infected people reduce contact with others the rapid growth of infected people gets severely reduced.

Keywords: SIR-model, Infectious novel diseases, Preventive measures, Community immunity, Epidemiology

1. Introduction

The world is becoming more and more connected through improvements in transportation. This increase in far-travel also elevates the risk of an epidemic spreading around the globe. When this outbreak becomes a pandemic it eventually impacts many small communities when an infected person enters these communities. A local government can try several preventive measures when a disease is introduced, but it is hard to know what effects they have. This research aims to better understand the effectiveness of preventive measures on a small community.

1.1. SIR-Model

The SIR-Model is a mathematical model that estimates the course of an epidemic in a population. It divides a population in 3 categories, these categories are susceptible people, infected people, and immune people. The amount of people is constant, no people die and none are born. A sick person infects a constant fraction of the healthy people. This fraction is modified by the basic transmission

coefficient, which is dependent on the ineffectiveness of the disease [3].

1.2. Preventive measures

In an enclosed system there are several strategies to slow down the extreme exponential growth of infected people during an outbreak. Potential solutions is encourage minimizing social contact or keep infected people together and be ware of healthy people interacting with them. With a very contagious disease this isolating of the infected cannot always be achieved so more drastic measures have to be taken. For instance partial or even general restrictions on meeting others all-together.

1.3. Scientific significance

Other models that are widely used are primarily models that predict an epidemic on a larger scale. These models can be divided into two categories, namely agent-based models and scale spatial meta population models. None of these different types of models are used to model an epidemic in a small community [1]. This paper attempts to fill in this gap by providing a new approach to disease modeling that is suitable for smaller populations by adding behaviour to the individuals.

2. Methodology

The model is a simple representation of an evenly distributed group with interaction between people at random intervals. No new people come or leave the simulated box. It is an extension of a classic box simulation where the behaviour of people is slightly more complex and the disease has a probability to infect on contact.

2.1. Rules

The people are randomly distributed and move with a random vector throughout the box. Whenever a healthy person touches a sick person there is a probability the healthy person gets infected too and in turn can infect others. At $T=0$ of the simulation a low number of people ($<1\%$ of the population) are sick. After a set amount of frames a sick

person becomes immune, meaning it is impossible to get sick by the disease again and is unable to give the disease to any other person. When there is no sick person left the simulation gets immediately stopped.

2.2. Disease

The disease is a novel virus, meaning that people have not encountered it before and therefore also don't have any build in immunity. The disease itself has a very short incubation period and people are able to transmit the disease shortly after they are infected.

2.3. Behaviour

This can be set differently per fraction of the population. A behaviour is divided into 3 different stages corresponding to the 3 different 'states' a person is in during spreading of a disease. With the option to move or stay in position during any one of these states.

2.4. Validating effectiveness

The effectiveness of the preventative measures is determined by the amount of stress that is put on hospitals, due to the limit capacity in hospitals [2]. This is directly related to the size of the peak % of infected people. A higher % of infected people means more hospitalized people and therefore more stress. Another important consideration is the risk of the epidemic reappearing again. To test the risk of the epidemic reappearing a low number of random people (<1% of the population) get infected and are able to spread it to other healthy people that are still left in the box. If the population is immune or if the measures are effective, then we expect only a small number of new cases to come up.

3. Results

In the box simulation the following behaviours are imitated: no behavioural change, infected people have less social contact, a portion cannot interact with the others. In addition the plot of a secondary outbreak on the last simulation is discussed. Before the second outbreak behaviour is changed to no change when they become infected. The main focus will lie on the peak of sick people and the amount of people being immune at the end of the simulation.

3.1. Normal spread

High susceptibility in the model result in these numbers of people per category in population over time. Much social contact with other people and long lasting infectiousness will increase the spread over a population.

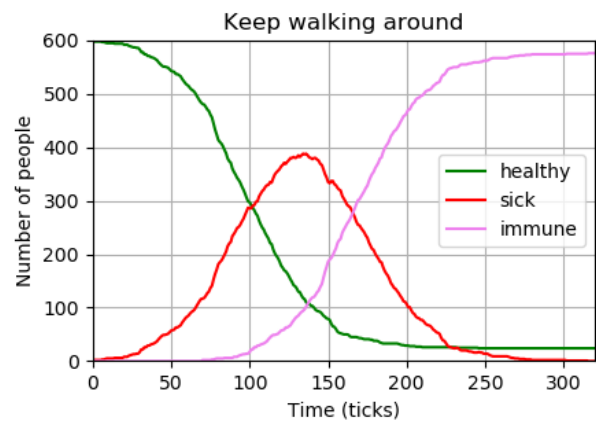


Figure 1: Healthy people follow a S-curve stabilising with only a small percentage of the population left. The amount of sick people follow a symmetrical distribution which peaks just under 66%. When around 25% of the population is immune the amount of sick people decreases. 96.00% of the population becomes immune

3.2. Less social contact when sick

Because the convergence of healthy people to immune takes much time there is a high percentage of people sick at once. The population is unable to have significant amount of immune people to lower the amount of sick people infecting healthy people before almost all healthy people are infected (this starts happening when <20% of the population is healthy).

Notable is the timeline being twice as long as in 3.1. When behaviour of the people is changed to an infected person standing still this person does not meet as many people during that period. In addition the sick people with create hot-spots where there are dense groups of sick people close together like in a hospital. Because they do not meet many new people gives the population time to increase the percentage of immunity. The chance a healthy person enters this group infected people gets smaller because it is a lower fraction of the population. Slowing down the spread of the disease gives time to get enough people immune.

3.3. Half the population does not interact

Even though the time frame is the same as 3.2 the end result is very different. Having half the population stand still overall decreases the frequency of contact between people. When a person got sick it stopped moving which makes them unable to infect half the population. This makes spreading the disease very difficult and so the amount of

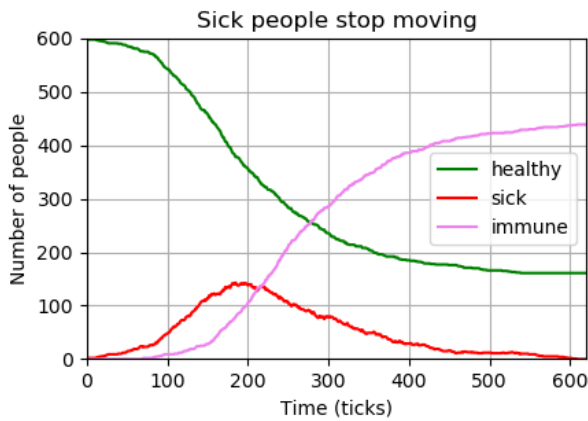


Figure 2: Healthy people follow a wide S-curve and stabilise. The amount of sick people follow a right skewed distribution which peaks just under 25%. When around 17% of the population is immune the amount of sick people decreases slowly. 73.16% of the population becomes immune

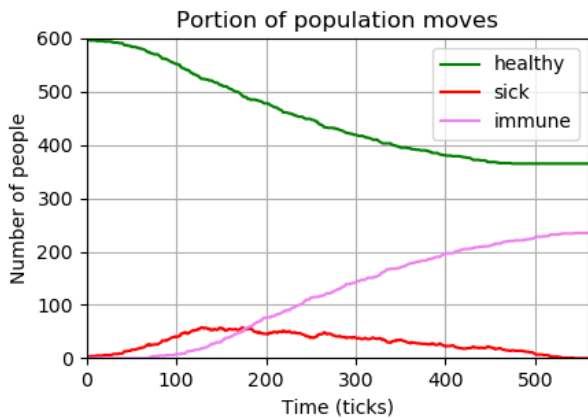


Figure 3: The amount of healthy people decrease very slowly and never surpass the amount of people who have become immune. The amount of sick people is right skewed with a linear slope going down. 39.17% of the population becomes immune

people that need to be immune to put a halt to the spread is also lower.

Although this method results in the lowest amount of people infected with the disease, this method does not make the whole population resistant all-together. Even though in the small population the disease has been eliminated, a

second portion of infected people during normal behaviour will still result in the population having a new outbreak.

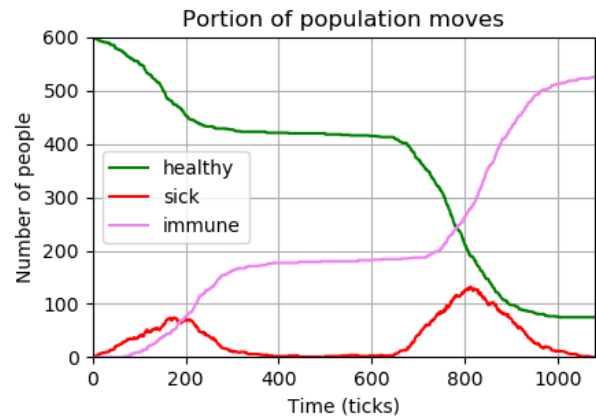


Figure 4: The first outbreak is similar to 3.3 but instead with 30.00% of the population immune. When around $T=600$ a small portion is infected again. This immediately results in a quick grow of infected people including a higher peak at one point of time. The system stabilises at 87.50% immunity

Whenever the fraction of immune people is too low a population is very susceptible to a secondary outbreak. This can be the result of an infected person coming inside a community from an area where the disease is still present. Simulating this scenario with the behaviour set to all people moving around again there is a second outbreak which results in an even greater peak of infected people. There are in the end less people immune than in 3.1 before the disease is gone again.

4. Discussion

4.1. Bias

In the simulation there is no bias where an individual will only meet a certain group of people and occasionally someone new. Every combination of two people have an equal chance to interact. To correct for this the simulation should include a preference to stay close to the location where they are at the start of the simulation.

4.2. Behaviour

The behaviour of the people in the simulation does not represent the true behaviour during the different states of a disease. In the model people will interact with a person even if they are sick whilst this is often not the case. A person will be sick and isolate itself from the society by

staying home from work, not going to school, and other group activities.

4.3. Differences between diseases

Diseases do not always require direct contact with another person, for example a disease can float through the air in a room or enter the groundwater and infect someone. In addition to this a disease can also be transmitted during specific portions of when people are sick which is not included in this model. A disease may not show symptoms for a fraction of the infected people so the behaviour of those does not change at all during spread.

4.4. Infecting

In reality diseases have a difference in chance to infect on social interaction. Outside where it is colder and UV-rays break the DNA/RNA structures to prevent the cell from being able to reproduce when entering a human body [4]. There is an equal probability to get infected at any location in the model.

5. Conclusion

Results show that only having a portion of the population interact with others has most effect on reducing the maximum amount of sick people at a single point. This reduces stress put on hospitals at the peak of an epidemic. The community is however still at risk of secondary outbreaks when local measures are eventually canceled.

The first and second model are not at risk from the disease reappearing again. The peak in the first model is almost a fraction of 66% and will therefore put significant stress on the health sector. So it is preferable to take a measure like model 2 with a peak of around 25%.

Even though this research aims to understand effects of preventive measures in disease spreading on a small scale. The effect is expected to be similar for bigger population groups.

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